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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 8

Title: End-to-End Business Analytics Capstone Project

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Experiment No.: 8

End-to-End Business Analytics Capstone Project

Problem Statement

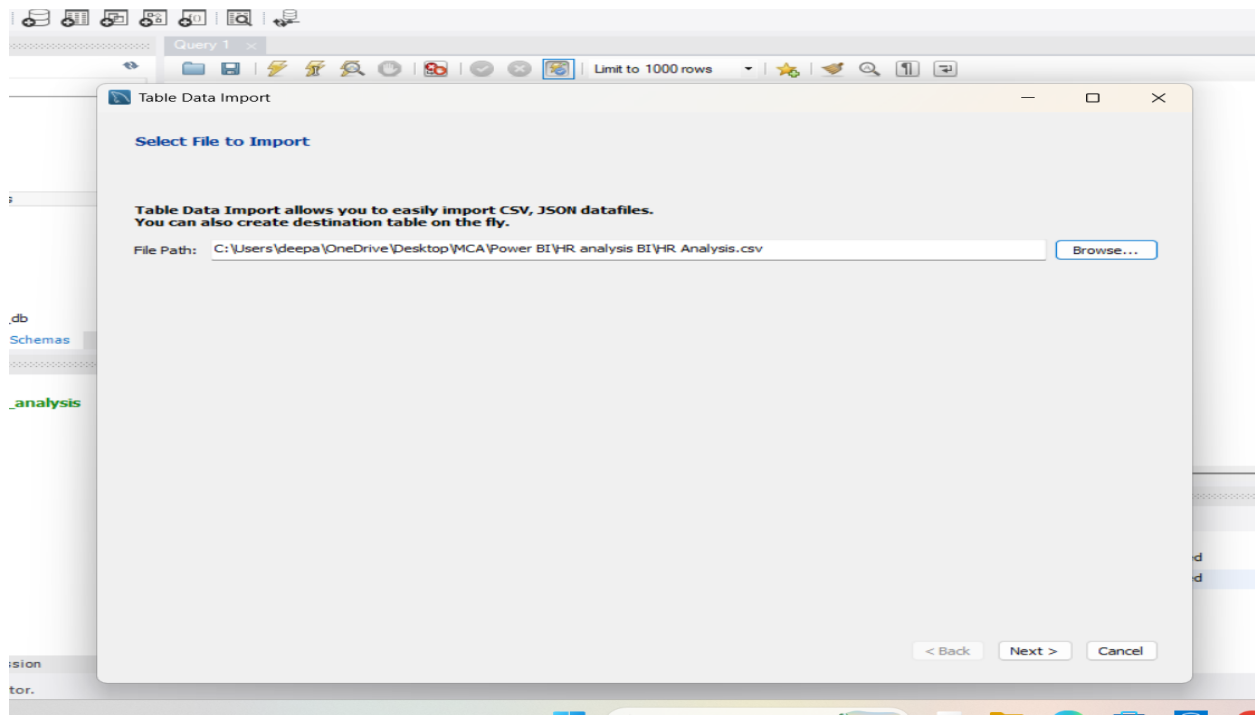
You are a Business Intelligence Consultant for a growing organization. Your task is to develop an end-to-end Business Analytics solution using a dataset of your choice to support data-driven business decisions.

Methodology / Steps Followed

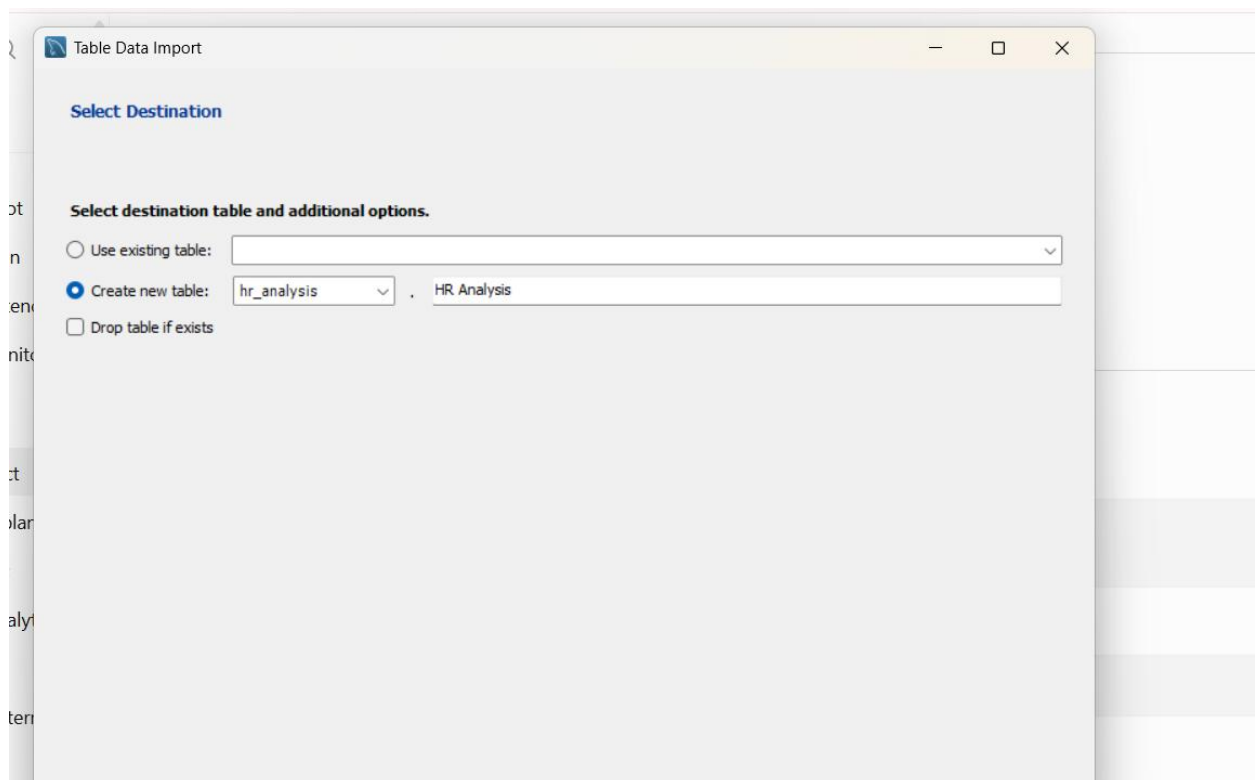
1. **Dataset Collection** – Collected the HR Analytics dataset containing employee demographics, job details, salary, satisfaction, performance, overtime, and attrition information.
2. **Data Storage in MySQL** – Imported the dataset into **MySQL Workbench** and created the **hr_analysis** database.
3. **Data Cleaning & Preparation** – Checked missing values, data types, duplicate records, and inconsistent column values using **MySQL and Power Query**.
4. **SQL Data Preparation** – Used SQL queries to inspect, filter, and prepare employee data for analysis.
5. **Data Modeling** – Created **Fact and Dimension tables** and established relationships using a **Star Schema** in Power BI.
6. **Power Query Transformation** – Performed data type conversion, column transformations, and creation of analytical categories such as **Age Group** and **Income Band**.
7. **DAX Measures & KPIs** – Created measures such as **Total Employees, Employees Left, Attrition Rate, Average Monthly Income, Job Satisfaction, and Overtime Employees**.
8. **Dashboard Development** – Developed interactive Power BI dashboards with KPI cards, charts, slicers, and a **Reset Filters** button.
9. **Business Analysis** – Analyzed attrition based on department, job role, income, satisfaction, overtime, work-life balance, performance, travel, distance, and tenure.
10. **Insights & Recommendations** – Identified key employee-retention patterns and provided actionable recommendations for improving workforce retention.
11. **Power BI Service Deployment** – Published the completed report to **Power BI Service** and configured an **On-premises Data Gateway** for MySQL connectivity.

Output:

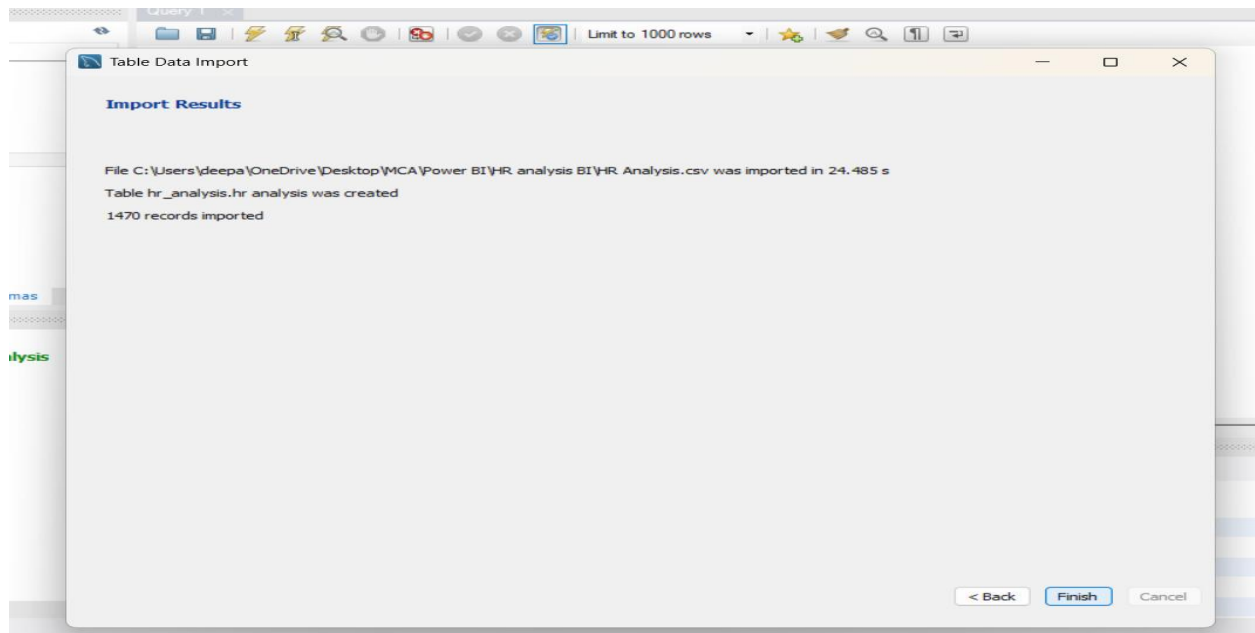
Importing Dataset into MySQL database:



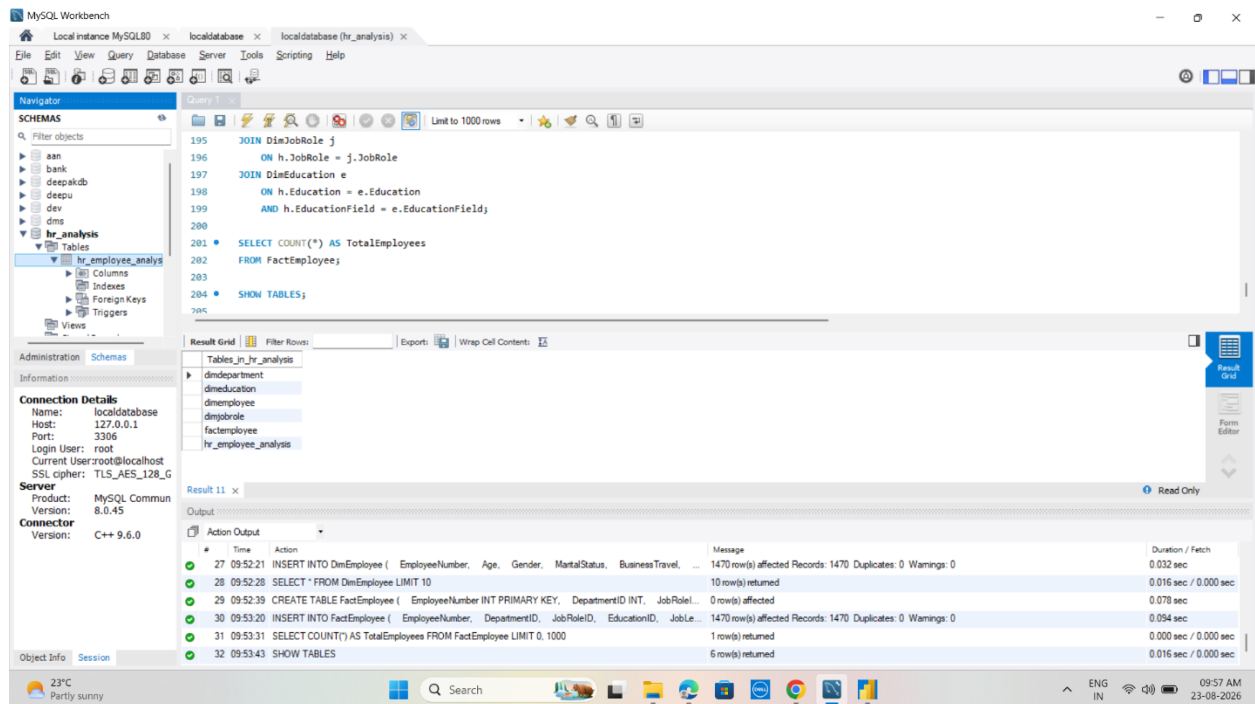
Creating Table:



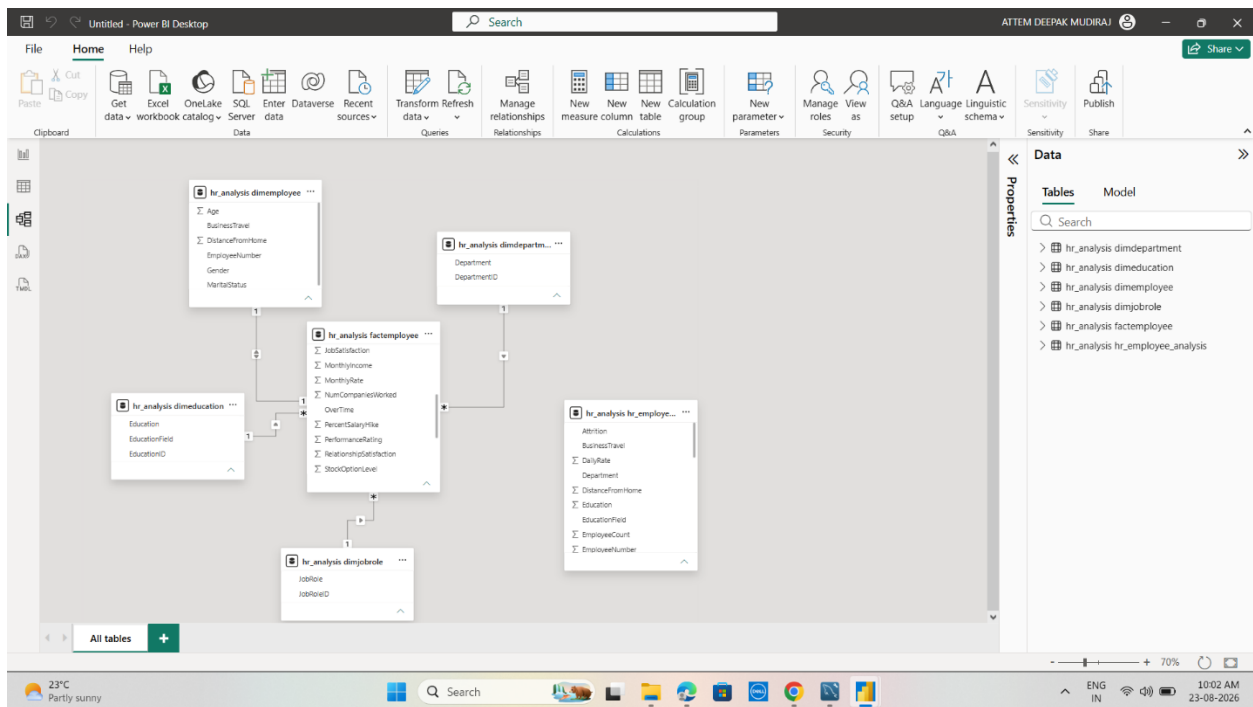
Imported Dataset:



Created Fact & Dimensions tables:



Model Relationships:



Using DAX to create New Measures:

The screenshot shows the Power BI Desktop interface with the 'Measure tools' ribbon active. A new measure named 'Employees Left' has been created using the following DAX formula:

```
1 Employees Left =  
2 CALCULATE(  
3 [Total Employees],  
4 'hr_analysis FactEmployee'[Attrition] = "Yes"  
5 )
```

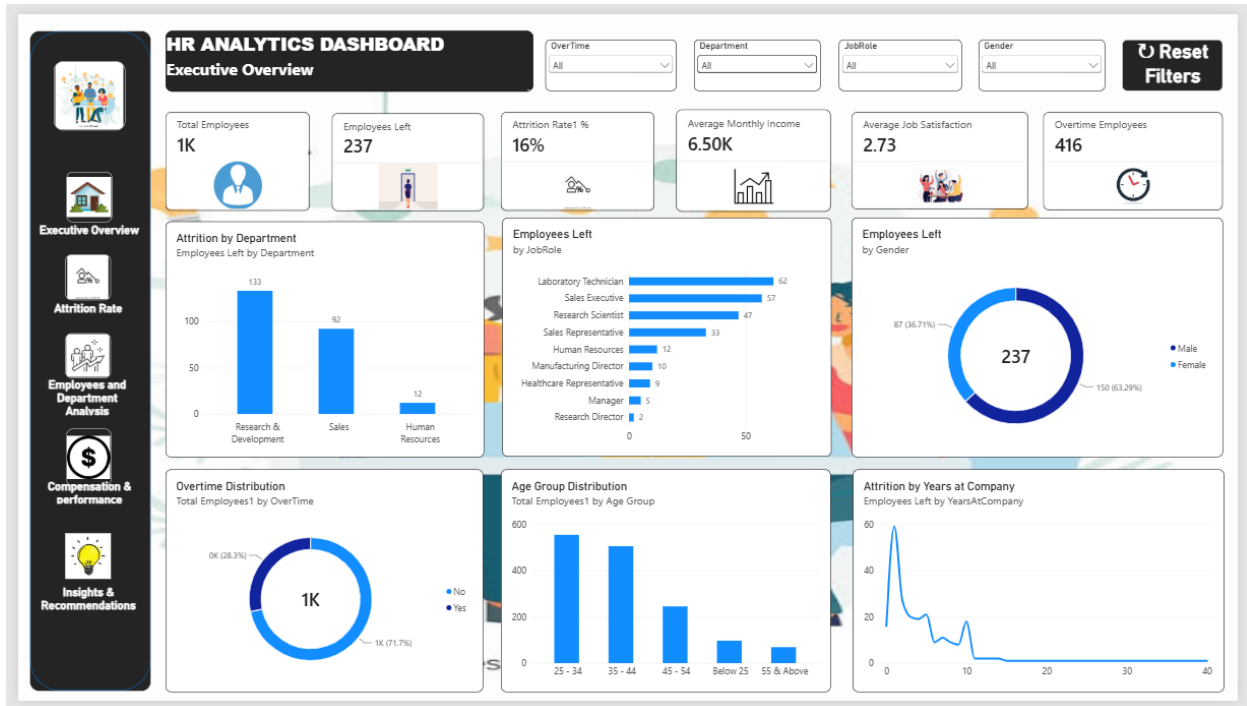
The table below displays the results of this measure, showing the number of employees who have attrited (left) for each row in the 'hr_analysis hr_employee_analysis' table. The table has 1,470 rows and 10 columns.

Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction
22	No	Non-Travel	7123	Research & Development	16	2	Medical	1	22	
39	Yes	Travel_Rarely	895	Sales	5	3	Technical Degree	1	42	
35	No	Travel_Rarely	464	Research & Development	4	2	Other	1	53	
27	No	Travel_Rarely	1240	Research & Development	2	4	Life Sciences	1	54	
48	Yes	Travel_Rarely	626	Research & Development	1	2	Life Sciences	1	64	
45	No	Travel_Rarely	1339	Research & Development	7	3	Life Sciences	1	86	
36	Yes	Travel_Rarely	318	Research & Development	9	3	Medical	1	90	
36	No	Travel_Rarely	132	Research & Development	6	3	Life Sciences	1	97	
32	No	Travel_Rarely	827	Research & Development	1	7	Life Sciences	1	134	
37	No	Non-Travel	1040	Research & Development	2	2	Life Sciences	1	139	
34	No	Travel_Rarely	1031	Research & Development	6	4	Life Sciences	1	151	
19	No	Travel_Rarely	7181	Research & Development	3	1	Medical	1	201	
51	No	Travel_Rarely	7169	Research & Development	7	4	Medical	1	211	
38	No	Travel_Rarely	1261	Research & Development	2	4	Life Sciences	1	271	
38	Yes	Travel_Rarely	7180	Research & Development	29	7	Medical	1	282	
41	No	Travel_Rarely	896	Sales	6	3	Life Sciences	1	298	
59	No	Travel_Rarely	142	Research & Development	3	3	Life Sciences	1	309	
41	No	Travel_Rarely	1411	Research & Development	19	2	Life Sciences	1	334	
37	Yes	Travel_Frequently	504	Research & Development	10	3	Medical	1	342	
51	No	Travel_Rarely	833	Research & Development	1	3	Life Sciences	1	353	
36	No	Travel_Frequently	566	Research & Development	18	4	Life Sciences	1	407	
39	Yes	Travel_Rarely	7162	Sales	3	2	Medical	1	445	
43	No	Travel_Rarely	1001	Research & Development	7	3	Life Sciences	1	451	
42	No	Travel_Rarely	810	Research & Development	21	5	Life Sciences	1	468	

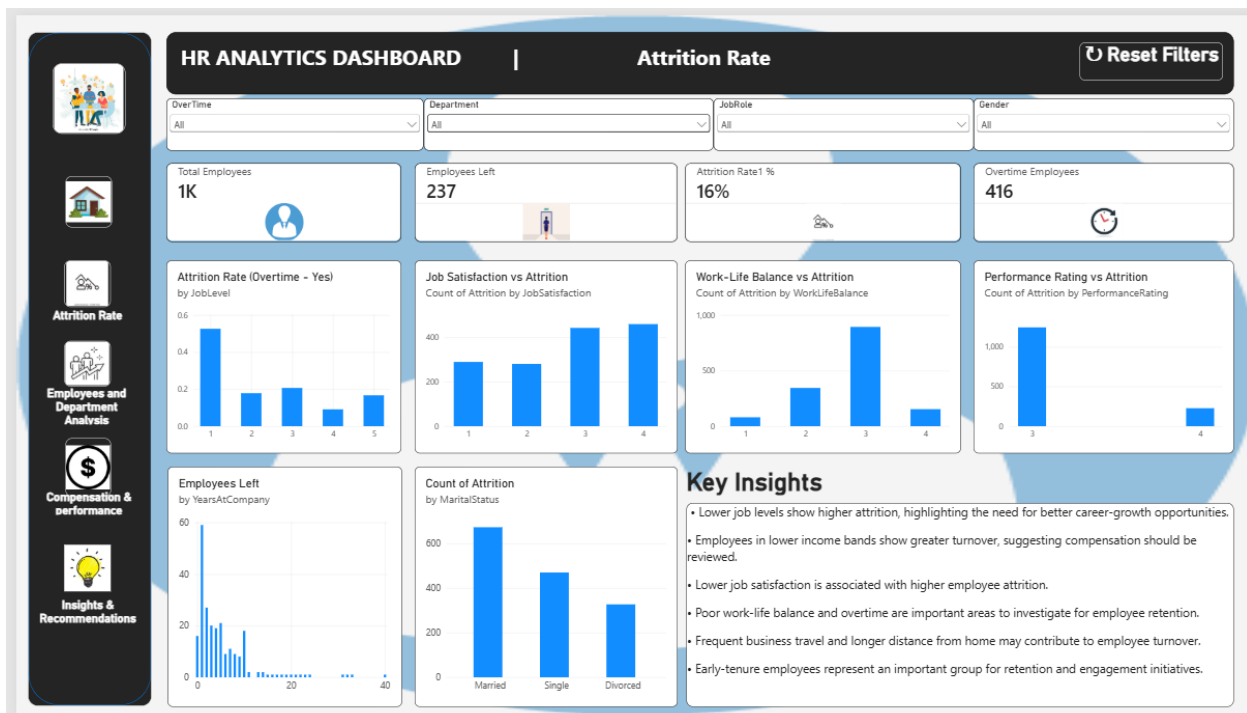
The bottom status bar indicates the table is 'hr_analysis hr_employee_analysis' with 1,470 rows and the column 'Employees Left' (0 distinct values).

Dashboard:

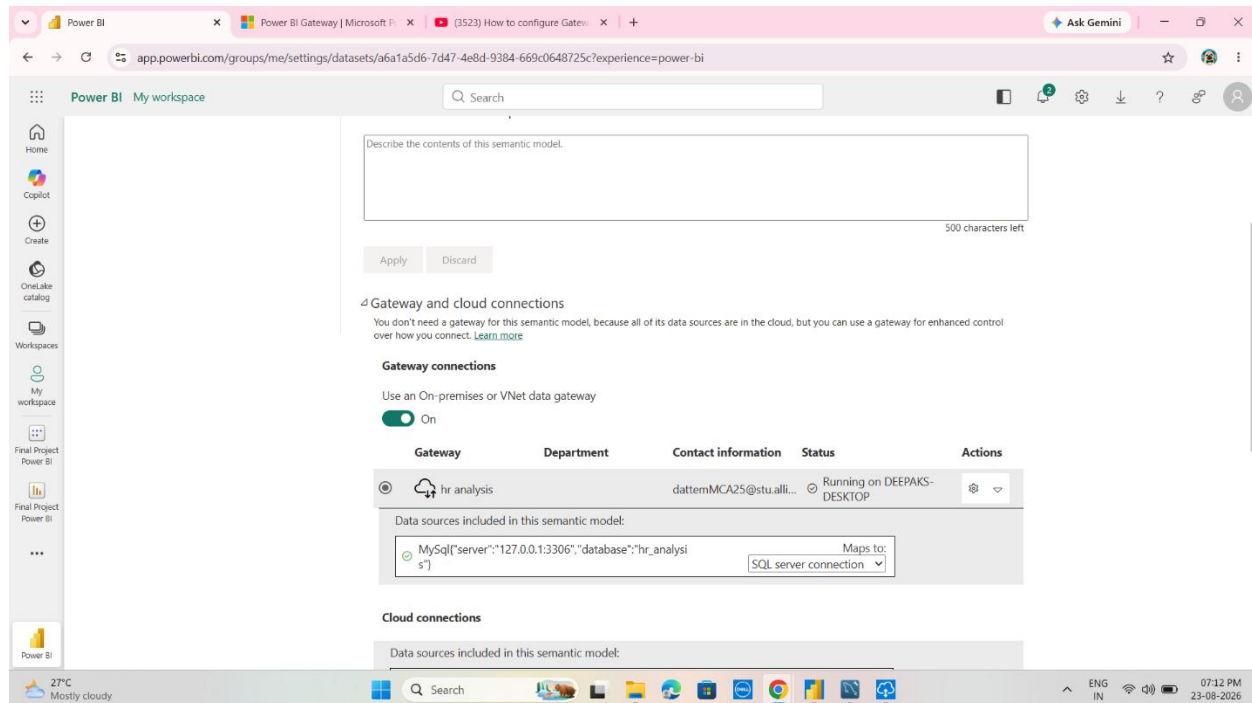
Page 1:



Page 2:



Connecting Power BI service:



The Final Power BI Dashboard Provides:

- **Interactive HR KPIs** such as total employees, employees who left, attrition rate, average income, and overtime employees.
- **Attrition analysis** by department, job role, gender, job level, and age group.
- **Employee satisfaction analysis** based on job satisfaction and work-life balance.
- **Compensation analysis** using monthly income and income bands.
- **Performance analysis** based on performance ratings and salary hikes.
- **Workforce behavior analysis** covering overtime, business travel, distance from home, and tenure.
- **Interactive filters and slicers** for dynamic exploration of employee data.
- **Reset Filters functionality** for quickly returning the dashboard to its default view.
- **Business insights and recommendations** to support employee retention and HR decision-making.
- **Power BI Service deployment** with MySQL connectivity through an on-premises data gateway.

Conclusion

The HR Analytics project successfully developed an **end-to-end Business Intelligence solution** by integrating MySQL with Power BI. The project transformed raw HR data into a structured **star-schema data model**, performed data cleaning and transformation using Power Query, and created meaningful **DAX-based KPIs and interactive dashboards**.

The analysis provided insights into **employee attrition, job satisfaction, compensation, overtime, work-life balance, performance, and workforce characteristics**. These insights can help HR managers identify potential retention risks and make more informed workforce decisions.

Overall, the project demonstrates the complete journey from **raw data → SQL database → data modeling → Power BI analysis → visualization → business insights**, providing a practical and data-driven approach to HR decision-making.

Learning Outcomes

1. Gained practical experience in **end-to-end Business Intelligence and Analytics**.
2. Learned to connect and manage **MySQL databases with Power BI**.
3. Developed skills in **SQL queries and data preparation**.
4. Learned **Power Query** for data cleaning and transformation.
5. Gained understanding of **Fact and Dimension tables and Star Schema modeling**.
6. Developed **DAX measures and KPI calculations** for business analysis.
7. Improved skills in designing **interactive Power BI dashboards** with slicers, filters, and bookmarks.
8. Learned to analyze HR data and identify **employee attrition and workforce trends**.
9. Developed the ability to convert analytical findings into **business insights and actionable recommendations**.
10. Gained experience in **Power BI Service deployment and data gateway connectivity**.
11. Improved overall understanding of how **data-driven analytics supports business decision-making**.

GitHub Repository: <https://github.com/deepakMudiraj06/Power-BI/tree/main>

GitHub Repository Folder: <https://github.com/deepakMudiraj06/Power-BI/tree/main/Final%20Project>